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Section A

1. what is data?

— Data is a collection of raw facts, value, information that can be stored and processed

2. what is Database.

— Database is an organized collection of data that can be accessed, stored, managed

3. what ~~is it~~ does DBMS stand for

— DBMS- Database management system & update efficiently

4) what does RDBMS stand for.

— RDBMS- Relational Database management System

5) what is MySQL?

— MySQL is an open source relational database management system that uses SQL to create relational database

6) what is SQL

— SQL is language used to communicate with relational database

7) Names any two ^{database} modules.

→ ① Network Database ② Relational Database

8) what is row in database table

9) what is the purpose of the DISTINCT keyword
 → Distinct keyword is a show the unique data

10) what does the % wildcard represent in MySQL LIKE
 — select * from student
 when name LIKE 'A.%';

Section B -

11) Differentiate between Database & RDBMS.
 — Database -

RDBMS -

12) Differentiate between SQL & MySQL.

SQL - SQL is a language

- used to write query

- contain SQL command

MySQL - MySQL is a software

- manage relational database

- execute SQL Statement

13) write any four SQL command categories & give one command for each

→ ① DDL - Data definition language

• create

• insert

• alter

and
it

IRE

2) DML - Data manipulation language

- Insert ✓
- update ✓
- Delete ✓

3) TCL - Transaction control language

- commit ✓
- Rollback ✓
- Save point ✓

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4) DQL - Data query language

- Select

5) DCL - Data control language

- Grant

14) what is the difference between IN & BETWEEN operators

IN - multiple possible values for a column

BETWEEN - BETWEEN a certain range

2

15) Explain IS NULL & IS NOT NULL with suitable SQL e.g

?

Section - C

16) explain DBMS & RDBMS write a least three difference betⁿ them

1) DBMS - Database management system

- General database management system

- Support different database depending on system

2) RDBMS - Relational Database management system

- type of DBMS

- Based on relational model

5

17) explain the following database with basic structure

1) Hierarchical database

→ Hierarchical database is data in tree-like structure

parent
|
+--- child

1

2) Network database

- Network database is interconnected record and multiple support

student
/ \
course teacher

1

3) Relational database

- Relational database is data into tables & rows and columns

1

#	stud_id	Name	age
	101	Vedika	20

4) object-oriented databases.

- object-oriented database is a store data in the form of object

student object

```

{
  name = "Vedika"
  age = 20
  course = "python"
}
  
```

5) NoSQL database

- NoSQL Database that use data models other than the traditional table.

(18) explain the six MySQL constraints give the notes with purpose

- NOT NULL - That column cannot have a NULL value.
- UNIQUE - That all values in column are different.
- DEFAULT - That automatically value provide.
- CHECK - That all satisfies special condition.
- FOREIGN KEY - relationship two tables & value ensure exists.
- PRIMARY KEY - NULL value and must be unique.

19) consider the following table

a) display students whose age is 20.

select * from student
where age = 20;

O/P

id	name	age	gender	city
1	Rahul	20	male	Delhi
6	Anjali	20	female	Jaipur

b) Display students whose age is between 20 & 30

select * from student
where age BETWEEN 20 and 30;

O/P

id	name	age	gender	city
1	Rahul	20	male	Delhi
6	Anjali	20	female	Jaipur

c) display student from Delhi or Mumbai using IN

select * from student
where city IN ("Delhi", "Mumbai");

O/P

id	name	age	gender	city
1	Rahul	20	male	Delhi
2	Pritya	21	female	Mumbai
8	Manish	25	male	Delhi

d] Display only female students

select * from student
where gender = "female";

o/p

id	name	age	gender	city
2	Pritya Rahul	21	female	mumbai
4	Neha	19	female	Gurgaon
6	Anjali	20	female	Jaipur
7	Pooja	21	female	Kolkata

e) display students whose name start with p
select * from student
where name LIKE "P%";

o/p

id	name	age	gender	city
2	Pritya	21	female	mumbai
7	Pooja	21	female	Kolkata

20) explain the mysql LIKE operator & its wildcards & write sql queries for

a) names starting with A.

select * from student
where name LIKE "A%";

o/p

id	name	age	gender	city
3	Amit	22	male	pune
6	Anjali	20	female	Jaipur

b) names ending with a
— select * from student
where name LIKE "l.a";

o/p	id	name	age	gender	city
	2	Priya	21	female	Mumbai
	4	Neha	19	female	Gurgaon
	7	Pooja	21	female	Kolkata

c) names containing am
— Select * from student
where name LIKE "l.aml";

o/p	id	name	age	gender	city
	1	Rahul	20	male	Delhi
	2	Priya	21	female	Mumbai
	3	Amul	22	male	Patna
	4	Neha	19	female	Gurgaon
	6	Anjali	20	female	Jalpur
	7	Pooja	21	female	Kolkata
	8	Manish	25	male	Delhi

d) names having a as the second character
— Select * from student
where name LIKE "al%";

o/p	id	name	age	gender	city
	1	Rahul	20	male	Delhi
	8	manish	25	male	Delhi

e.g. - names having 0 as the third character.

— select * from student
when name LIKE "_0%";

Pooja

21] explain the following mysql concepts with syntax & e.g.

a) DISTINCT

— select DISTINCT column-name
from table-name;

e.g. - select DISTINCT (city)
from student;

b) LIMIT

— Syntax - select * from table name
LIMIT 10;

e.g. select * from student
LIMIT 10;

c) LIMIT with OFFSET

— select * from table name
LIMIT 10, 5;

eg

d) to COUNT()

Syntax - select * from table name

select COUNT (column name)
from table name;

e.g. select COUNT (*) from student;

e. SUM()

Syntax -

select SUM (column name)
from table name;

e.g. select SUM (age)
from student;